STAT2004 – Tutorial Week 10

We will cover Exercises 1 and 2 together in the tutorial. Exercise 3 is optional. Please complete Exercises 4–6 for homework, and check your solutions with your tutor next week.*

Exercise 1 (Likelihood ratio test for exponential data)

A manufacturer advertises that 90% of their LED light bulbs last over 24 months before failure. To test if this claim is an exaggeration, one light bulb from this manufacturer is randomly selected and tested, and its time-to-failure X is recorded.

The time-to-failure of a product can often be modelled by an exponential distribution, so let $X \sim \exp(\lambda)$, with pdf and cdf given respectively by

$$f_\lambda(x) = \lambda e^{-\lambda x} \quad \text{and} \quad F_\lambda(x) = 1 - e^{-\lambda x} \ , \quad x \leq 0 \ ,$$

where $\lambda > 0$ is an unknown parameter, often interpreted as an *instantaneous rate of failure*.

(a) Convert the manufacturer's claim into a null hypothesis H_0 about λ. Also, write down the alternative hypothesis H_1.

(b) Construct the likelihood ratio test for testing H_0 against H_1 based on the datum X, and show that this reduces to the test with rejection region

$$\{X \leq d \ , \quad \text{for some critical value } d \ \} \ .$$

Does this test make intuitive sense?

(c) Find the critical value d so that the Type I error rate of this test is controlled at 5%.

(d) Is the test from parts (b) and (c) the uniformly most powerful test at the 5% level for this scenario? Briefly explain why, or why not.

Suppose that after 12 months and 12 days of usage, the light bulb died.

(e) Based on the observed outcome, would you accept or reject the manufacturer's claim?

(f) Supplement your decision by computing and interpreting a p-value.

Exercise 2 (Testing the precision of an apparatus)

In engineering, the *precision* of a random process is defined as $1/\sigma^2$, where σ^2 is its variance. Higher precision is therefore equivalent to smaller variance, and vice versa. It is important in engineering to test whether a certain random process is producing output with an acceptable level of precision. This is equivalent to testing the variance.

A machine in the physics lab that measures gravity has been correctly calibrated, so that measurements coming from this machine follow a normal distribution with mean $\mu = 9.807\text{m/s}^2$ and some variance σ^2. Suppose that the industry-standard for the allowable variance of such machines is σ_0^2. To investigate whether this particular machine meets industry requirements, we make n independent measurements, $X_1, X_2, \ldots, X_n$, on this machine at random times over a period of three days.

(a) Write down the null and alternative hypotheses in terms of the parameter σ^2.
(b) Write down the likelihood function $L(\sigma^2|\boldsymbol{X})$ for σ^2 given the data $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)^\top$.
(c) Construct the likelihood ratio test for this problem, and show that it is equivalent to the test with rejection region

$$\left\{ \sum_{i=1}^{n} (X_i - 9.807)^2 \geq c \ , \quad \text{for some critical value } c \right\} .$$

(d) What is the distribution of the test statistic $\sum_{i=1}^{n} (X_i - 9.807)^2$ under the null?
(e) Explain how the critical value c should be chosen so that the Type I error rate is 5%.

Now, suppose that we are also unsure about the calibration of this machine, so that both μ and σ^2 are unknown. A natural modification of our above test is to replace the assumed mean of 9.807 with the sample mean, $\bar{X}$, leading to the test with rejection region

$$\left\{ \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2 \geq d \ , \quad \text{for some critical value } d \right\} . \tag{1}$$

(f) What is the distribution of the test statistic $\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2$ under the null?
(g) Explain how the critical value d should be chosen so that the Type I error rate is 5%.
(h) Suppose that the industry-standard is $\sigma_0^2 = 0.02^2$ and that $n = 18$ samples are to be taken. Write out the decision rule explicitly for this scenario.

A machine exhibiting a bias can be re-calibrated by a lab technician. However, if the precision of a machine is not up to industry standard, then this is a problem with the manufacturing process and the machine needs to be sent back to the manufacturer. Thus, testing a new lab machine often involves the following sequential process:

1. First, test whether the machine meets industry standard in terms of precision using rule (1). If the machine fails this test, it is immediately sent back to the manufacturer.
2. If the machine passes this test (that is, $\sigma^2 = \sigma_0^2$), then test the mean of the machine to see if it is properly calibrated.

(i) If both of these tests are carried out at the 5% level, what is the overall chance that a perfectly good machine will fail this test?
(j) If we want the overall Type I error rate of this sequential test to be 5%, at what level should the two individual tests be carried out?

Exercise 3* (Sum of Poisson random variables)

The Poisson distribution is extremely useful for modelling count responses over a period of time, space, volume, etc... Examples include modelling the number of enquiries to a call-centre over a given period of time, the number of bugs in a section of computer code, the number of pollen particles in some volume of air, and so on... This question investigates the distribution of the sum of independent Poisson random variables.

(a) Let y be any nonnegative integer. Show by induction (or otherwise) the binomial theorem:

$$(\lambda_1 + \lambda_2)^y = \sum_{j=0}^{y} \frac{y!}{(y-j)!\ j!}\ \lambda_1^{y-j}\ \lambda_2^j$$

(b) Let $X_1 \sim \text{Poisson}(\lambda_1)$ independently of $X_2 \sim \text{Poisson}(\lambda_2)$, and write $Y = X_1 + X_2$ for their sum. Argue why

$$P(Y = y) = \sum_{x_1=0}^{y} \Big[P(X_1 = x_1)\ P(X_2 = y - x_1) \Big] , \quad y = 0, 1, 2, \ldots$$

[*Aside*: the expression above is called a *convolution*.]

(c) Using parts (a) and (b), or otherwise, show that

$$P(Y = y) = \frac{(\lambda_1 + \lambda_2)^y}{y!}\ e^{-(\lambda_1+\lambda_2)} , \quad y = 0, 1, 2, \ldots$$

Hence, what is the distribution of Y?

(d) Now let $X_1, X_2, \ldots, X_n$ be an iid sample from a Poisson(λ) distribution. Using part (c), or otherwise, show that the total count $Y = X_1 + X_2 + \ldots + X_n$ follows a Poisson($n\lambda$) distribution.

Exercise 4 (Hypothesis tests for the Poisson rate)

Suppose that the number of enquiries coming into a call-centre follows a Poisson distribution. A call-centre employee is being paid on the assumption that the call rate is $\lambda_0 = 7.3$ calls per hour. The employee feels that she is being underpaid, i.e., she suspects that the true call rate is in fact higher than 7.3 calls per hour. To investigate this issue, the company randomly samples $n = 15$ one-hour blocks over a period of two weeks, and records the number of calls $X_1, X_2, \ldots, X_{15}$ received in each block.

(a) Write down the null and alternative hypotheses being tested in this scenario.

(b) Show that the MLE of the underlying rate λ is the same as the method-of-moments estimator, namely $\bar{X} = (X_1 + X_2 + \ldots + X_n)/n$.

(c) In light of part (b), a decision rule of the form $\bar{X} \geq c$ is proposed for this problem (i.e., "reject the null hypothesis if the sample mean number of calls is overly large"). How should we set the critical value c so that the Type I error is no more than 5%?
[*Hint*: The result from Exercise 1(d) states that the sum $X_1 + X_2 + \ldots + X_n$ of iid Poisson(λ) random variables has a Poisson($n\lambda$) distribution.]

(d) If the true call rate is in fact 9.0 calls per hour, what is the power of this test? Interpret the power in the form of a sentence.

(e) Plot the power curve of this test for true call rates between 7.3 and 11.0 calls per hour.

The observed number of calls received in the 15 randomly sampled one-hour blocks were:

7 11 10 11 6 7 10 9 7 5 11 5 9 6 18

(f) Decide between the two hypotheses from part (a) based on the observed data.

(g) Supplement your decision from part (f) by computing a p-value.

Exercise 5 (One-sided confidence intervals for the Poisson rate)

Assume the same scenario as in Exercise 5. As we demonstrated in the lectures, there is generally no pivot transformation for discrete data, such as Binomial or Poisson data. For large sample sizes we can appeal to the Central Limit Theorem to approximate the distribution of $\bar{X}$, however, with only $n = 15$ samples here, this is not justified. Instead, we can construct construct confidence intervals for the Poisson rate λ via inverting hypothesis tests.

Given the observed data as before,

$$7 \quad 11 \quad 10 \quad 11 \quad 6 \quad 7 \quad 10 \quad 9 \quad 7 \quad 5 \quad 11 \quad 5 \quad 9 \quad 6 \quad 18 \ ,$$

recall that in Exercise 5(f) and (g) we rejected the the null $H_0 : \lambda = 7.3$ in favour of the alternative $H_1 : \lambda > 7.3$ at the 5% level.

(a) Suppose that the null hypothesis value was $\lambda_0 = 7.5$ instead. Would we still reject $H_0 : \lambda = 7.5$ in favour of $H_1 : \lambda > 7.5$ at the 5% level? Answer this by finding the p-value for this test.

(b) Suppose that the null hypothesis value was $\lambda_0 = 7.6$ instead. Would we still reject $H_0 : \lambda = 7.6$ in favour of $H_1 : \lambda > 7.6$ at the 5% level? Answer this by finding the p-value for this test.

(c) Show that the largest λ_0 value that we would still reject at the 5% level is $\lambda_0 = 7.579$ (up to 3 decimal places).

Hence, any λ_0 value in the range $(7.580, \infty)$ would be accepted in the one-sided hypothesis test of $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda > \lambda_0$ at the 5% level. An equivalent way of saying this is that "we can be 95% confident that the true call rate is at least 7.58 calls per hour".

This is an example of a (right) one-sided 95% confidence interval procedure constructed via inverting hypothesis tests. In this case, the 95% confidence interval is precisely the set of all parameter values we would have retained as the null hypothesis in a test at the 5% level.

(d) Using analogous calculations to part (c), compute a 90% right one-sided confidence interval for λ based on the observed data. Interpret this confidence interval in the form of a sentence.

Exercise 6 (Power curves, sample size and relative effect size)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N(\mu, 1^2)$ distribution, where the population mean μ is an unknown parameter.

(a) Construct the likelihood ratio test of $H_0 : \mu = 0$ versus $H_1 : \mu = \mu_1 > 0$, and show that it reduces to the decision rule $\{\bar{X} \geq c\}$ for some critical value c.

(b) For sample size $n = 5$:

   i. find the value c such that the level of the test is $\alpha = 0.01$.

   ii. Graph the power of this test against alternative values μ_1 in the range $0 \leq \mu_1 \leq 2$.

(c) Repeat part (b) for sample sizes $n = 10, 20, 40$ and plot all the power curves on the same graph. You may use different colours for each sample size and include a legend/key to make the graph easier to read.

Now suppose $X_1, X_2, \ldots, X_n$ are a random sample from a $N(\mu, 10^2)$ distribution, where the population mean μ is an unknown parameter.

(d) Repeat parts (a)–(c), this time graphing the power curves alternative values μ_1 over the range $0 \leq \mu_1 \leq 20$.

(e) What do you notice about the graphs in part (c) and the graphs in part (d)?

We see that the power of this test depends on the **sample size** and the **relative effect size** $\delta = \mu_1/\sigma$ (also known as **signal-to-noise ratio**) rather than the absolute effect size μ_1.

(f) In our medical trial example, we have $\sigma = 1.1$ mmol/L as our day-to-day variability in LDL cholesterol levels. Suppose we suspect that the new drug can improve cholesterol levels by 0.6 mmol/L, on average. Using your graphs from the previous parts, what sample size do we need in order to correctly reject $H_0 : \mu = 0$ with power at least 80%?

This combination of

1. restricting the Type I error rate to $\alpha = 0.01$; and
2. aiming for power $\geq 80\%$ for correct detection of a postulated relative effect size $\delta = \mu_1/\sigma$

is commonly used as a guideline for setting the sample size in medical studies.